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BIO 539

Github Repository Link: <https://github.com/fleischerowen1/BIO-539-Final-Project>

BIO 539 Final Project

Introduction

To reach an understanding of an animal's ecophysiology, its fine-scale movement parameters, such as quantitative behavioral patterns and rate of energy expenditure, must first be described (Kadar et al., 2019). In large elasmobranchs and other marine predators with vast horizontal ranges, the necessary data for these descriptions are lacking (Bullock et al., 2015). Recent advancements in biologging technology have allowed researchers to improve the quantification of fine-scale movement and identify a wide range of behavioral states (Waller et al., 2023).

Tailbeat frequency (TBF) is one of the most commonly derived metrics of finescale movement in teleost and cartilaginous fishes. TBF describes the rate of oscillatory swimming movements and correlates tightly with speed, acceleration and energy expenditure. Due to its descriptive nature, TBF allows researchers to investigate how animals utilize energy during different behavioral states and in varying environmental conditions. In past studies, TBF has been used to estimate activity levels and uncover movement strategies in a variety of different fish species.

Yo-yo diving is one of the most commonly observed movement behaviors of large, pelagic marine animals, particularly piscivorous fish and consists of repeated excursions from the surface to great depth and then back to the surface (Tokunaga et. al, 2025). The depth profile of fish exhibiting yo-yo diving often resembles the dive profile of a marine mammal or any obligate air breather. There are many theories as to why large, pelagic fish would exhibit this yo-yo dive pattern since they are not constrained in their ability to respire. Some researchers suggest yo-yo diving increases search area to include different more of the water column, while others propose the behavior serves energetic or thermoregulatory functions.

In this analysis, shortfin mako shark TBF variation across four different dive stages; descent, ascent, deep horizontal swimming and surface swimming, was investigated. We

hypothesize that each dive phase will have distinct energetic demand on the shark, resulting in a significant variation of TBF.

Methods

Data collection

Accelerometer data was collected from one male shortfin mako shark caught off Rhode Island, USA in October of 2021. The tag package contained an accelerometer (recorded tri-axial acceleration at 100 Hz), a gyroscope (recorded tri-axial angular acceleration at 100 Hz), a depth/temperature reader (recorded depth/temp at 1 Hz), as well as a camera and an acoustic receiver for tag package recovery. 24 hr galvanic time releases kept the tag package attached to the dorsal fin of the shark.

Data processing

Before analysis, data was pre-processed in IGOR Pro 9 software to extract movement metrics. Continuous wavelet transformation was applied to the gyroscope derived signals to extract tailbeat amplitude (TBA) and tailbeat cycle (TBC). Once the pre-processed dataset was in R, the datasets were processed using the tidyverse, dplyr and diveMove packages.

Depth and timestamp data were imported into R and cleaned to remove incomplete observations and converted into POSIXct format. This conversion allows for the synchronization of datasets and the creation of a time-depth recorder (TDR) object. A TDR object is required by the diveMove package to characterize the diving behavior, and once created, can apply these characteristics to each individual dive. Individual dives were separated into three phases; descent, ascent and bottom phase (deep horizontal swimming). Depth recordings from 0-4m were classified as “surface swimming”. Once the depth and timestamp data were annotated with the subsequent dive stage for every row of data, this data was merged into a single dataframe with the tailbeat data.

Statistical analysis

Differences in tailbeat frequency among dive stages were evaluated using a one-way analysis of variance (ANOVA). Following identification of significant overall differences, Tukey post-hoc comparisons were conducted to determine which dive stages differed significantly from one another. Statistical significance was evaluated at $\alpha = 0.05$.

The proportion of time spent within each dive stage was also calculated by summarizing the number of observations assigned to each behavioral category. Data visualization was performed using ggplot2, including bar plots and boxplots to compare the distribution and central tendency of tailbeat frequency across dive stages.

Results

Dive classification identified four primary behavioral stages throughout the deployment track: descending, ascending, deep horizontal swimming, and surface swimming. The proportion of time spent in each dive stage varied across the track (Fig. 1), with surface swimming and deep horizontal swimming being the two main dive stages

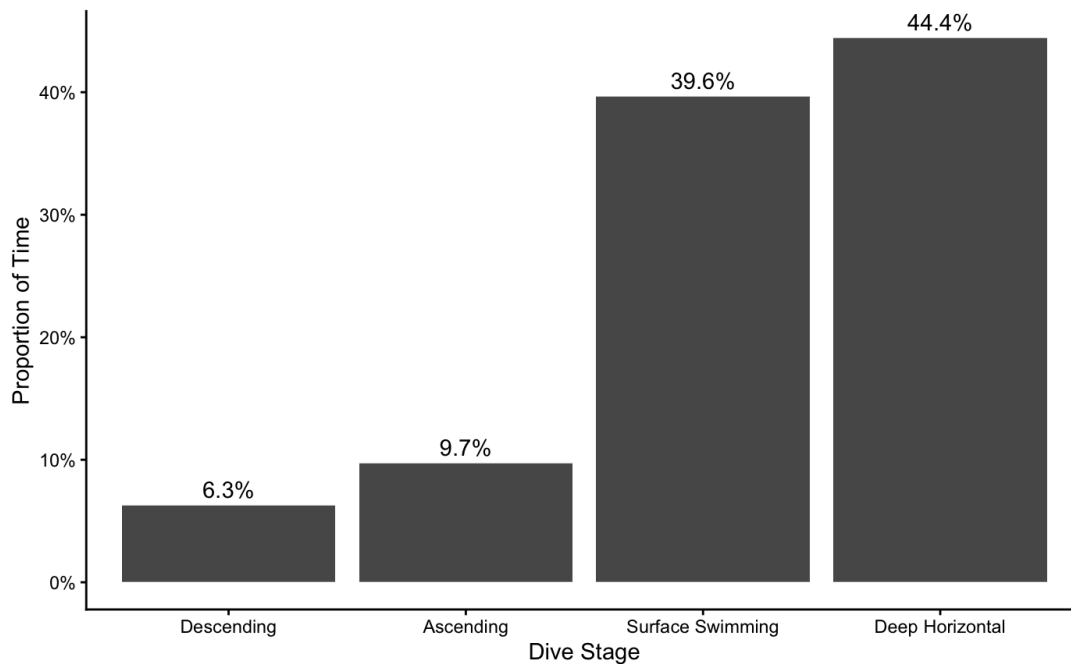


Figure 1: Proportion of total time spent in each dive stage during the track of a single shortfin mako shark (*Isurus oxyrinchus*). Dive behavior was categorized into four different behaviors; surface swimming, descending, ascending, and deep horizontal swimming phases using depth-based classifications generated in the diveMove package in Rstudio. Values above bars represent the percentage of total observations assigned to each behavioral category.

Tailbeat frequency (TBF) differed significantly across dive stages ($p < 0.001$). Post-hoc Tukey comparisons revealed that all comparisons among dive stages were significantly different ($p < 0.001$). Ascending dives exhibited the highest tailbeat frequencies, while descending dives

displayed the lowest values. Surface swimming and deep horizontal swimming demonstrated intermediate TBF values.

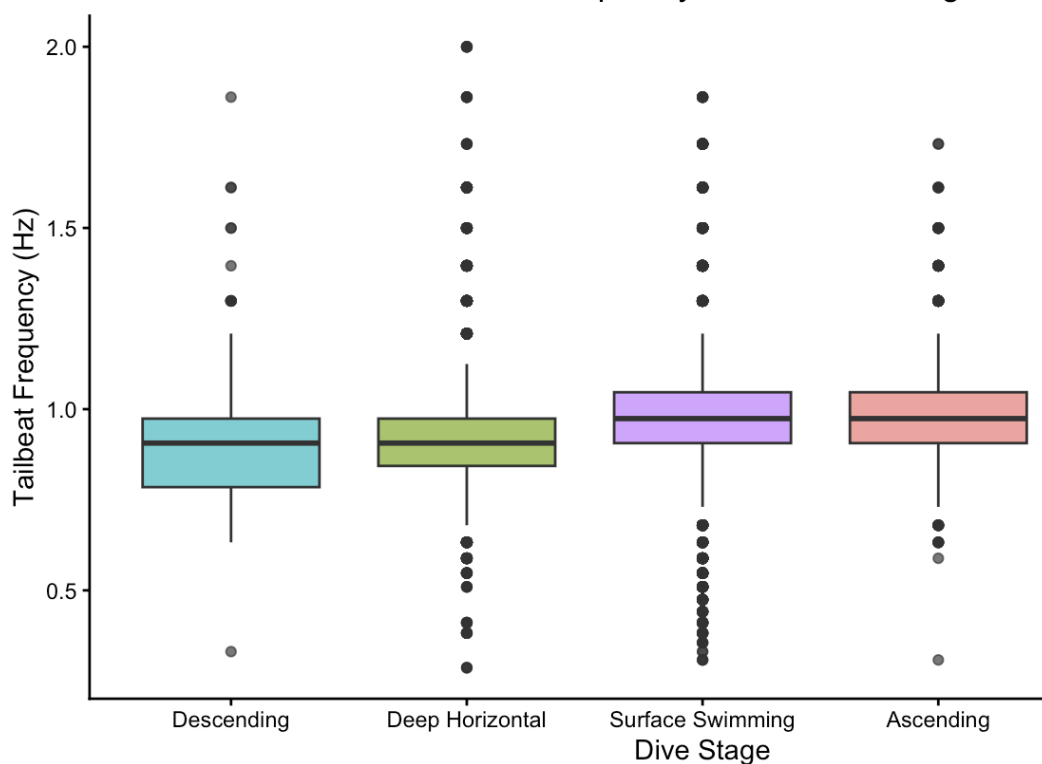


Figure 2: Distribution of tailbeat frequency (TBF) across four classified dive stages in a shortfin mako shark (*Isurus oxyrinchus*). Boxplots display the median, IQR, and overall variability of TBF values for descending, deep horizontal swimming, surface swimming, and ascending behaviors. TBF differed significantly among all dive stages (ANOVA with Tukey post-hoc comparisons, $p < 0.001$), with ascending behavior exhibiting the highest overall TBF values and descending behavior exhibiting the lowest.

Discussion

This analysis has shown that TBF differed significantly across dive stages in a shortfin mako shark and suggests that the energy expenditure changes depending on the subsequent dive stage. Of all the identified dive stages, ascents exhibited the highest TBF values while descending had the lowest. The findings presented support the hypothesis that different dive phases impose different energetic demands on sharks.

The increased TBF during ascent dive phases is likely due to the shark swimming against gravity, as shortfin mako sharks lack a swim bladder and must generate hydrodynamic lift to maintain their position in the water column. Conversely, the TBF values during descent the use of

gliding behavior, which can be achieved from a negative pitching of the body and allowing the negative buoyancy to pull the shark to depth.

References

- Bullock, R. W., Guttridge, T. L., Cowx, I. G., Elliott, M., & Gruber, S. H. (2015). The behaviour and recovery of juvenile lemon sharks *Negaprion brevirostris* in response to external accelerometer tag attachment. *Journal of Fish Biology*, 87(6), 1342–1354. <https://doi.org/10.1111/jfb.12808>
- Kadar, J., Ladds, M., Mourier, J., Day, J., & Brown, C. (2019). Acoustic accelerometry reveals diel activity patterns in premigratory Port Jackson sharks. *Ecology and Evolution*, 9(16), 8933–8944. <https://doi.org/10.1002/ece3.5323>
- Tokunaga, S., Chiang, W., Nakamura, I., Matsumoto, R., & Watanabe, Y. Y. (2025). Enhanced thermoregulation abilities of shortfin mako sharks as the key adaptive significance of regional endothermy in fishes. *Journal of Animal Ecology*. <https://doi.org/10.1111/1365-2656.70116>
- Waller, M. J., Queiroz, N., Ivo da Costa, Tiago Cidade, Loureiro, B., Womersley, F. C., Fontes, J., Afonso, P., Macena, B. C. L., Loveridge, A., Humphries, N. E., Southall, E. J., & Sims, D. W. (2023). Direct measurement of cruising and burst swimming speeds of the shortfin mako shark (*Isurus oxyrinchus*) with estimates of field metabolic rate. *Journal of Fish Biology*, 103(5), 864–883. <https://doi.org/10.1111/jfb.15475>